

IPv4 Addressing

1. OSI Layer 3 – Network Layer Review

Key Responsibilities of Layer 3

Function	Detail
Connectivity	Links end-hosts on DIFFERENT networks (beyond the local LAN)
Logical Addressing	Uses IP addresses (assigned by admin, not burned-in like MAC)
Path Selection	Chooses the best route across complex networks (e.g. the Internet)
Device	Routers operate at Layer 3

Layer 2 vs Layer 3 Addressing

	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	MAC Address	IP Address
Assignment	Burned-in at manufacture	Configured by administrator
Scope	Same LAN only	Different networks
Device	Switch	Router

Network Topology – How Routers Separate Networks

Switches **expand** a single network. A **Router** separates networks and controls traffic between them. Broadcasts do NOT cross routers.

Network 1 (192.168.1.0/24)		Router R1		Network 2 (192.168.2.0/24)
PC1 – 192.168.1.1 PC2 – 192.168.1.2	→	G0/0: .1.254 G0/1: .2.254	→	PC3 – 192.168.2.1 PC4 – 192.168.2.2
SW1		(separates broadcast domains)		SW2

■ Key Rule: Broadcasts are limited to the local network and do NOT cross routers.

2. IPv4 Address Structure

An IPv4 address is 32 bits (4 bytes), written in Dotted Decimal notation.

Bit Layout of 192.168.1.254

Octet 1	Octet 2	Octet 3	Octet 4
192	168	1	254
11000000	10101000	00000001	11111110

8 bits	8 bits	8 bits	8 bits
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Network vs Host Portion (Prefix Length)

Prefix	Network Bits	Host Bits	Network Portion	Host Portion	Class
/8	8	24	First 1 octet	Last 3 octets	A
/16	16	16	First 2 octets	Last 2 octets	B
/24	24	8	First 3 octets	Last 1 octet	C

3. Binary (Base 2) – Explained

Binary uses only **0** and **1**. Each bit position has double the value of the position to its right. An 8-bit octet can represent 0–255.

Bit Position Values in an Octet

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
128	64	32	16	8	4	2	1

Worked Examples – Binary ↔ Decimal

Decimal	Binary	Calculation
192	11000000	$128 + 64 = 192$
168	10101000	$128 + 32 + 8 = 168$
1	00000001	$1 = 1$
254	11111110	$128+64+32+16+8+4+2 = 254$
143	10001111	$128+8+4+2+1 = 143$
118	01110110	$64+32+16+4+2 = 118$
221	11011101	$128+64+16+8+4+1 = 221$
127	01111111	$64+32+16+8+4+2+1 = 127$
207	11001111	$128+64+8+4+2+1 = 207$

Decimal → Binary Conversion Method (Subtraction)

1. Write the 8 bit-position values: 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1
2. Starting from 128, subtract each value from your number if possible.
3. Write 1 if you subtracted; write 0 if you could not.
4. Continue until remainder = 0; fill remaining positions with 0.

Range: An 8-bit octet can represent values from 0 (00000000) to 255 (11111111).

4. IPv4 Address Classes

IPv4 addresses are divided into 5 classes based on the **first octet**. CCNA focuses on Classes A, B, and C.

Class	Leading Bits	First Octet Range	Prefix	Networks	Hosts/Network	Use
A	0	1 – 126	/8	Few (126)	Very many (~16M)	Unicast
B	10	128 – 191	/16	Medium (16K)	Medium (~65K)	Unicast
C	110	192 – 223	/24	Many (2M)	Few (254)	Unicast
D	1110	224 – 239	—	—	—	Multicast
E	1111	240 – 255	—	—	—	Experimental

First Octet Range – Visual Bar

1–126 Class A	127 Loopback	128–191 Class B	192–223 Class C	224–239 Class D	240–255 Class E
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■ ■ 127.0.0.0 – 127.255.255.255 is the Loopback range. Traffic sent here is processed back up the local TCP/IP stack (RTT = 0 ms). Therefore Class A is effectively 1–126, not 0–127.

5. Number Systems Recap

System	Base	Digits Used	Example (3294)
Decimal	10	0–9	3294
Hexadecimal	16	0–9, A–F (A=10...F=15)	CDE (C×256 + D×16 + E×1)
Binary	2	0, 1	110011011110

6. Quick-Reference Cheat Sheet

IPv4 length	32 bits = 4 octets = 4 bytes
Dotted decimal	Four decimal numbers separated by dots e.g. 192.168.1.1
Octet range	0 – 255 (8 bits: $2^8 = 256$ values)
/24 network	First 24 bits (3 octets) = network; last 8 bits = host
Switch vs Router	Switch = expands one LAN; Router = separates LANs
Broadcast domain	Routers block broadcasts; switches forward them
Loopback	127.x.x.x – loops back to local device, RTT = 0 ms
Class A default	/8 prefix – few networks, millions of hosts each
Class B default	/16 prefix – medium networks and hosts
Class C default	/24 prefix – many networks, 254 hosts each
Binary tip	Start from 128, subtract if possible, write 1; else write 0

IPv4 Addressing – Part 2

(continued) · Netmasks · Network & Broadcast Addresses · Binary

1. Address Class Size Summary

How many networks and hosts does each class support?

Class	Network Bits	Host Bits	Possible Networks	Hosts per Network	Usable Hosts
A	8	24	128	~16,777,216	16,777,214
B	16	16	~16,384	~65,536	65,534
C	24	8	~2,097,152	256	254

■ Usable hosts = (Total hosts) – 2 because the first address is the Network Address and the last is the Broadcast Address — neither can be assigned to a device.

Relative Host Capacity per Class (visual scale)



2. Subnet Mask (Netmask)

Cisco devices use a **dotted-decimal netmask** instead of slash notation. The network portion is all **1s**; the host portion is all **0s**.

Class	Prefix	Binary Netmask	Dotted Decimal Netmask
A	/8	11111111.00000000.00000000.00000000	255.0.0.0
B	/16	11111111.11111111.00000000.00000000	255.255.0.0
C	/24	11111111.11111111.11111111.00000000	255.255.255.0

■ /24 and 255.255.255.0 are the same thing — just different notations. Slash notation (CIDR) is used on Juniper; dotted-decimal mask is used on Cisco.

3. Network Address & Broadcast Address

For the network **192.168.1.0 /24 (Class C)**:

Address	Value	Host Portion	Assignable?	Role
Network Address	192.168.1.0	All 0s (00000000)	NO	Identifies the network

First Usable Host	192.168.1.1	00000001	YES	e.g. PC1
... (other hosts)	192.168.1.2 – .253	00000010–11111101	YES	Any host
Last Usable Host	192.168.1.254	11111110	YES	e.g. Router G0/0
Broadcast Address	192.168.1.255	All 1s (11111111)	NO	Send to ALL hosts
Network Address Host portion = all 0s First address in the subnet Cannot be assigned to a host Identifies the network itself		Broadcast Address Host portion = all 1s Last address in the subnet Cannot be assigned to a host Sends traffic to ALL hosts		

■ When a packet is sent to the broadcast address (e.g. 192.168.1.255), the destination MAC in the Ethernet frame is FF:FF:FF:FF:FF:FF (all Fs — Layer 2 broadcast).

4. Day 7 – Full Topic Summary

Dotted Decimal & Binary	IPv4 = 32 bits; written in dotted decimal (4 octets) for human readability.
Network & Host Portion	Hosts on the same network share the same network portion but have unique host portions.
IPv4 Address Classes	A (1–126 /8), B (128–191 /16), C (192–223 /24), D (Multicast), E (Experimental).
Prefix Length / Netmask	Two ways to express the same thing: /24 $\equiv$ 255.255.255.0
Network Address	Host bits all 0s; first address; identifies the subnet; NOT assignable.
Broadcast Address	Host bits all 1s; last address; sends to all hosts on the LAN; NOT assignable.
Loopback Range	127.0.0.0/8 — traffic loops back locally; RTT = 0 ms; not routed.

5. Quiz – Binary ↔ Dotted Decimal Practice

Binary → Dotted Decimal (Q1–Q5) and Dotted Decimal → Binary (Q6–Q10)

Q	Direction	Input	Answer
1	Bin → Dec	00111111.00111000.11100111.00010011	63.56.231.19
2	Bin → Dec	11110011.01111111.01100010.00000001	243.127.98.1
3	Bin → Dec	01101111.00000110.01011001.11000111	111.6.89.199
4	Bin → Dec	11001111.11000110.00101111.01001100	207.198.47.76
5	Bin → Dec	01100100.11001001.00100001.11111101	100.201.33.253
6	Dec → Bin	88.46.90.91	01011000.00101110.01011010.01011011

7	Dec → Bin	221.234.246.163	11011101.11101010.11110110.10100011
8	Dec → Bin	3.41.143.222	00000011.00101001.10001111.11011110
9	Dec → Bin	10.200.231.91	00001010.11001000.11100111.01011011
10	Dec → Bin	248.87.255.152	11111000.01010111.11111111.10011000

6. Binary Conversion Reference Card

Bit-position values for a single octet:

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
128	64	32	16	8	4	2	1

Common decimal ↔ binary quick-lookup:

Dec	Bin	Dec	Bin	Dec	Bin	Dec	Bin
0	00000000	1	00000001	2	00000010	4	00000100
8	00001000	16	00010000	32	00100000	64	01000000
128	10000000	192	11000000	224	11100000	240	11110000
248	11111000	252	11111100	254	11111110	255	11111111

CCNA 200-301

IPv4 Addressing – Part 2

Day 8 · Host Calculation · Cisco IP Configuration · Show Commands

Jeremy's IT Lab · Free CCNA Course

1. IPv4 Address Classes – Quick Review

■ Class A range is technically 0–127, but 0.x.x.x and 127.x.x.x are both reserved. The usable/practical range is **1–126**. Always remember both facts for the exam.

Class	Leading Bits	First Octet Range	Usable Range	Prefix	Host Bits
A	0	0–127	1–126	/8	24
B	10	128–191	128–191	/16	16
C	110	192–223	192–223	/24	8
D	1110	224–239	(Multicast)	—	—

E	1111	240–255	(Experimental)	—	—
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2. Calculating Hosts, Network & Broadcast Addresses

The Formula

$$\text{Usable Hosts} = 2^N - 2$$

Where **N** = number of host bits

Subtract 2 to remove the **Network Address** (host bits all 0s) and **Broadcast Address** (host bits all 1s).

Worked Examples – All Three Classes

Class	Network	Prefix	Host Bits	Total Addr	Usable Hosts	Formula
C	192.168.1.0	/24	8	256	254	$2^8 - 2$
B	172.16.0.0	/16	16	65,536	65,534	$2^{16} - 2$
A	10.0.0.0	/8	24	16,777,216	16,777,214	$2^{24} - 2$

Finding Network, Broadcast, First & Last Usable Addresses

Address Type	Rule	Class C Example 192.168.1.0/24	Class B Example 172.16.0.0/16	Class A Example 10.0.0.0/8
Network Address	Host bits ALL 0s	192.168.1.0	172.16.0.0	10.0.0.0
First Usable Host	Network Address + 1	192.168.1.1	172.16.0.1	10.0.0.1
Last Usable Host	Broadcast Address – 1	192.168.1.254	172.16.255.254	10.255.255.254
Broadcast Address	Host bits ALL 1s	192.168.1.255	172.16.255.255	10.255.255.255

■ Quick trick: Network address = keep network octets, set host octets to 0. Broadcast address = keep network octets, set host octets to 255.

3. Configuring IP Addresses on Cisco Routers

Lab Network – R1 with 3 Interfaces

Interface	Network	Class	IP Assigned to R1	Subnet Mask	PC IP
G0/0	10.0.0.0/8	A	10.255.255.254	255.0.0.0	10.0.0.1 (PC1)
G0/1	172.16.0.0/16	B	172.16.255.254	255.255.0.0	172.16.0.1 (PC2)
G0/2	192.168.0.0/24	C	192.168.0.254	255.255.255.0	192.168.0.1 (PC3)

Step-by-Step Configuration Commands

Step 1 – Enter Privileged Exec Mode	<pre>R1> enable R1#</pre>
Step 2 – Enter Global Config Mode	<pre>R1# configure terminal R1(config)#</pre>
Step 3 – Enter Interface Config Mode	<pre>R1(config)# interface gigabitethernet0/0 R1(config-if)# Shortcuts: int g0/0 int GigabitEthernet0/0 (no space needed)</pre>
Step 4 – Assign IP Address	<pre>R1(config-if)# ip address 10.255.255.254 255.0.0.0 Shortcut: R1(config-if)# add 10.255.255.254 255.0.0.0</pre>
Step 5 – Enable the Interface	<pre>R1(config-if)# no shutdown Shortcut: no shut ✓ You will see two messages: "GigabitEthernet0/0 changed state to up" → Layer 1 "Line protocol on GigabitEthernet0/0 changed state to up" → Layer 2</pre>
Step 6 – Switch Directly to Next Interface (no need to exit)	<pre>R1(config-if)# interface gigabitethernet0/1 R1(config-if)# ip address 172.16.255.254 255.255.0.0 R1(config-if)# no shut</pre>

4. Essential "show" Commands

show ip interface brief — Column Breakdown

Column	What it Shows	Notes
Interface	Name of each network interface	e.g. GigabitEthernet0/0
IP-Address	Assigned IP address	"unassigned" if none set

OK?	Whether IP is valid (legacy field)	Should always be YES on modern devices
Method	How IP was assigned	"manual" = CLI configured; "unset" = none
Status	Layer 1 physical status	"up" = cable + enabled; "administratively down" = shutdown command active
Protocol	Layer 2 data-link status	"up" only if Status is also up

■ Router interfaces are **administratively down by default** (shutdown applied). Switch interfaces are NOT — they come up automatically when a cable is connected.

Three Key Show Commands

Command	Shortcut	Purpose
show ip interface brief	sh ip int br	Quick overview: IP, status, protocol for all interfaces
show interfaces <name>	sh int g0/0	Detailed Layer 1 & 2 info, MAC address (BIA), IP address
show interfaces description	sh int desc	Lists interface descriptions — useful for documenting topology

■ **BIA** = Burned-In Address = the permanent hardware MAC address of the interface. A different MAC can be configured via CLI but is rarely needed.

Interface Description Command

```
R1(config-if)# description ## Connected to SW1 ##
```

- Descriptions are optional but highly recommended for documentation.
- No strict format rules; use whatever makes the network clear.
- Shortcut: desc ## Connected to SW1 ##

5. Quiz – Network / Host Address Calculations

For each IP address, find: Network Address · Max Hosts · Broadcast Address · First Usable · Last Usable

Q	IP / Prefix	Class	Network Address	Max Hosts	Broadcast Address	First Usable	Last Usable
1	43.109.23.12 /8	A	43.0.0.0	16,777,214	43.255.255.255	43.0.0.1	43.255.255.254
2	129.221.23.13 /16	B	129.221.0.0	65,534	129.221.255.255	129.221.0.1	129.221.255.254
3	209.211.3.22 /24	C	209.211.3.0	254	209.211.3.255	209.211.3.1	209.211.3.254
4	2.71.209.233 /8	A	2.0.0.0	16,777,214	2.255.255.255	2.0.0.1	2.255.255.254
5	155.200.201.141 /16	B	155.200.0.0	65,534	155.200.255.255	155.200.0.1	155.200.255.254

■ Formula reminder: Max Hosts = $2^N - 2$ where N = host bits. Class A → N=24 → 16,777,214 | Class B → N=16 → 65,534 | Class C → N=8 → 254

Quick-Reference

Usable hosts formula	$2^N - 2$ (N = host bits)
Network Address	Host bits all 0s → first address, NOT assignable
Broadcast Address	Host bits all 1s → last address, NOT assignable
First Usable	Network Address + 1
Last Usable	Broadcast Address - 1
Class A usable range	1-126 (0.x.x.x and 127.x.x.x are reserved)
Enter interface mode	interface g0/0 or int g0/0
Assign IP	ip address e.g. ip address 10.1.1.1 255.0.0.0
Enable interface	no shutdown (no shut)
Router default state	Interfaces are administratively down – must use no shutdown
Switch default state	Interfaces come UP automatically when cable connected
Check all interfaces	show ip interface brief
Check one interface	show interfaces g0/0
Check descriptions	show interfaces description
Add description	description ## text ##
